

Model Paper - Database Management Systems

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Q1 (20 marks)

Consider the following Enhanced Entity-Relationship

(EER) diagram: I

Figure: EER



a) Convert the following EER model into the

relational model. In

b) What is the best option for mapping the ISA

hierarchies in the

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the

following set of fun

a) Find all the keys in relation R using the

attribute closure m

b) Is R in 3NF? Give reasons for your conclusion. (3

marks)

c) Is R in BCNF? Give reasons for your conclusion.

If R is not in BCNF

Q3 (25 marks)

A university is implementing a database system to

manage its studie

a) Write SQL queries to in type 2 driver, jdbc api

calls are converted

b) Code Segment: ```java String sql = "SELECT * FROM`

employees WHER

c) Briefly explain the options available for

transferring data b

d) Briefly explain how authentication and

authorization are a

e) A university is establishing role-based access

control for its da

i. Write a T-SQL statement to create a login to

Sarah with window

Q4 (40 marks)

Consider the following schema of a database designed

for a University:

a) Write SQL Queries to perform the following:

i. Find the names of students who are enrolled in

the course titled 'D

b) b) Create a function to calculate the total fee

amount for a given

c) c) Create a trigger that automatically updates

the 'amount' column